GeoLink: A 3D-aware Framework to Improve Generalization for Cross-View Geo-Localization

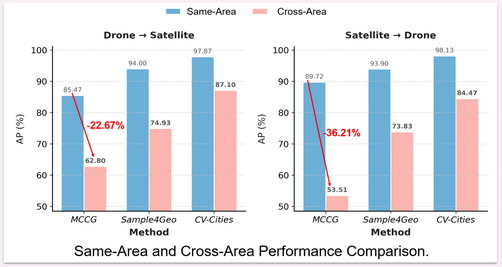
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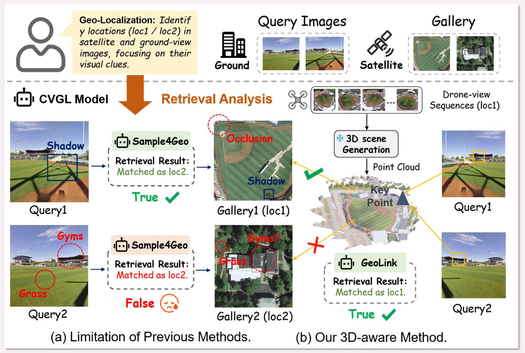
Problem



Same-Area and Cross-Area Performance Comparison.

Challenge: *Semantic inconsistency caused by view changes (e.g., Occlusion) , which further leads to poor cross-area performance of previous method.*

Motivation



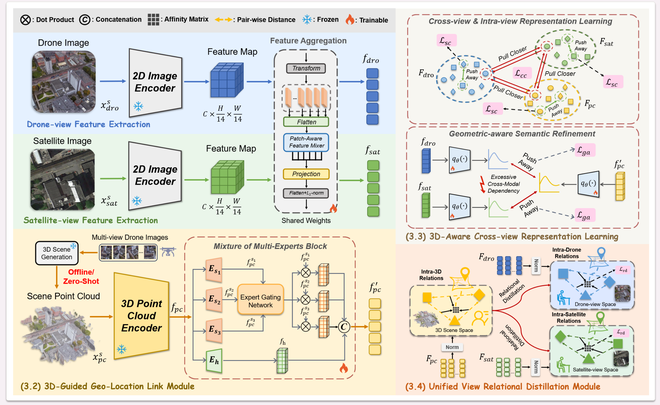
(a) Limitation of Previous Methods. (b) Our 3D-aware Method.

Occlusion **Limited FoV** **Domain shift**

- **2D-based methods are vulnerable to occlusion, ambiguous visual cues.**
- Scene-level 3D structures constructed from VGGT offer a comprehensive anchor for learning transferable visual information.

Method

1. **3D-Guided Geo-Location Link Module.** Reconstructs 3D geometry from frozen VGGT and adaptively fuses local and global 3D structural features through multi-expert gating.
2. **3D-aware cross-view representation learning.** Combines cross-view and intra-view contrastive learning with mutual information regularization to eliminate 2D-3D redundancy.
3. **Unified View Relation Distillation Module.** Distills relations from 3D teacher features into 2D students through knowledge distillation, improving the effectiveness of cross-view alignment.



Overall Optimization

$$\mathcal{L}_{total} = \mathcal{L}_{cc} + \mathcal{L}_{rd} + \mathcal{L}_{ga} + \lambda \mathcal{L}_{sc}$$

where λ is the weighting factor that balances the contribution of intra-view contrastive learning (set to 4.0).

Experimental Results

☐ Cross-View Image Retrieval

METHOD	R@1	AP
Sample4Geo	82.03	85.12
CV-Cities	90.27	91.77
QDFL	92.01	92.82
GeoLink	93.44	94.38

+1.43 R@1 and +1.56 AP over the prior SOTA on University-1652→SUES-200.

SO WHAT → *3D-aware training improves transfer across both geographic and adverse-weather domain shifts.*

↑ 2.45% R@1 · DenseUAV→U1652

↑ 3.58% R@1 · DenseUAV→SUES

☐ Multi-weather Robustness Evaluation

METHOD	R@1	AP
Fog-Rain	81.78	84.59
Fog-Snow	78.34	81.53
Rain-Snow	89.03	90.70
Wind	88.50	90.43
Avg.	84.41	86.81

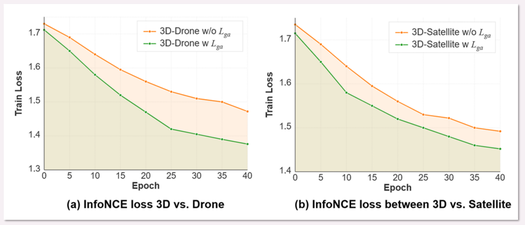
+4.22 R@1 and +3.40 AP over the prior SOTA on University-1652→SUES-200 (Multi-weather).

2D-Only Inference

90.50M parameters · **91.72** GFLOPs, comparable to Baseline CV-Cities (90.50M / 91.67G).

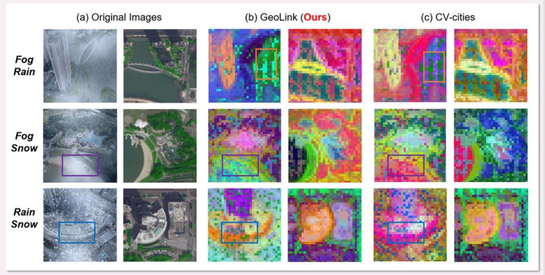
Ablation Study

SETTING	SUES→DUAV
Baseline → w/ MME	R@1 · **15.56** → **17.86**
All losses · 3 experts	**33.03 R@1 · 23.12 AP**



(a) InfoNCE loss 3D vs. Drone (b) InfoNCE loss between 3D vs. Satellite

Visualization Analysis



Takeaway

Beyond 2D visual matching: GeoLink suppresses view-biased redundancy and transfers 3D structural relations into 2D features, improving cross-area and adverse-weather performance while keeping inference image-only.

2D-only inference **Zero-shot transfer** **Multi-weather robust**